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ACM Transactions on Mathematical Software

Special Issue on Mixed-Precisions Techniques for HPC Algorithms and Applications using Hardware Accelerators

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In Memoriam: Stanimire Tomov

This special issue honors the memory of **Stanimire Tomov**, an exceptional researcher whose foundational contributions to numerical linear algebra, GPU computing, and mixed-precision algorithms have had a lasting impact on the high-performance computing community. As a major contributor to the **MAGMA** project and a leading advocate for leveraging hardware accelerators in mathematical software and had an impactful influence on their use in scientific applications, Stan helped bridging rigorous numerical analysis with high-performance implementations on emerging architectures. Equally important, he was a generous collaborator and mentor; his vision and dedication continue to inspire progress in mixed-precision methods and accelerator-driven HPC.

Aims and Scope

Over the past decade, mixed-precision numerical algorithms have become central to high-performance computing, propelled by rapid advances in both hardware accelerators—GPUs, tensor cores, FPGAs, and increasingly AI-oriented architectures—and new methods exploiting proliferation of floating-point data representations. By combining precisions strategically for arithmetic and/or storage, these approaches can deliver substantial improvements in performance, energy efficiency, and memory footprint, while preserving mathematically sound accuracy guarantees.

This special issue of **ACM Transactions on Mathematical Software (TOMS)** focuses on mixed-precision algorithms, software, and performance models for HPC applications, with particular emphasis on hardware accelerators and production-quality scientific computing. We invite original, high-quality submissions that advance the theory, design, implementation, and evaluation of mixed-precision techniques in mathematical software and large-scale scientific applications.

Topics of Interest

Topics of interest include, but are not limited to:

- Mixed-Precision Algorithms
 - Mixed-precision iteration and refinement accompanied by robust error correction techniques
 - Adaptive precision algorithms with accuracy/performance trade-offs
 - Convergence analysis and error bounds in reduced- and mixed-precision arithmetic

- Fixed-point, block floating-point, and other low-precision formats (FP16, BF16, FP8, custom formats)
- Validation and benchmarking of mixed-precision algorithms using emulated arithmetic and adaptation of formats not directly related to floating-point
- Linear Algebra and Numerical Kernels
 - Dense and sparse linear algebra in mixed-precision
 - Eigenvalue, SVD, and matrix factorizations using reduced precision
 - Krylov methods and solvers leveraging mixed-precision
 - Communication-avoiding, bandwidth-aware, and latency-mitigating methods for mixed-precision algorithms
- Hardware Accelerators and Architectures
 - Exploiting GPUs, tensor/matrix cores, and AI accelerators for mixed-precision computing
 - Performance modeling of mixed-precision kernels on heterogeneous systems
 - Energy efficiency and memory bandwidth optimization via reduced precision
 - Mixed-precision on emerging architectures (AI chips, FPGAs, chiplets, CGRAs, near-memory computing)
- Mathematical Software and Libraries
 - Design and implementation of mixed-precision software libraries
 - Extensions to existing HPC libraries (e.g., BLAS, LAPACK, MAGMA, SLATE, PETSc, Ginkgo, GraphBLAS)
 - Portability, reproducibility, and numerical robustness in mixed-precision software
 - Auto-tuning, runtime adaptation, and precision selection frameworks
- Applications and Case Studies
 - Mixed-precision in large-scale scientific simulations
 - Applications in AI-enabled scientific computing
 - Domain-specific applications (e.g., climate, seismic imaging, bioinformatics, materials science)
 - End-to-end performance and accuracy studies on leadership-class systems

Important Dates (tentative)

- Submissions deadline: June 1st 2026
- First-round review decisions: July 15th 2026
- Deadline for revision submissions: September 15th 2026
- Notification of final decisions: October 15th 2026
- Tentative publication: December 15th 2026

Submission Guidelines

Submissions must present original research that has not been published or simultaneously submitted elsewhere. All manuscripts should be prepared according to the ACM Transactions on Mathematical Software author guidelines. Papers will undergo the journal's standard rigorous peer-review process. Authors are encouraged to emphasize software design, reproducibility, numerical analysis, and practical impact, in line with the scope of TOMS.

Submission Procedure

Manuscripts should be submitted via the ACM TOMS submission system. During submission, authors must select the article type corresponding to this Special Issue on *Mixed-Precisions for HPC Algorithms and Applications using Hardware Accelerators*. The cover letter should explicitly state that the manuscript is submitted for consideration

in this special issue. All accepted papers will be published as part of the special issue and appear online individually as soon as they are ready. Detailed formatting instructions, templates, and submission policies can be found on the official ACM Transactions on Mathematical Software website.

For questions and further information, please contact **Jack Dongarra** (dongarra@icl.utk.edu), **Hatem Ltaief** (hatem.ltaief@kaust.edu.sa) and **Piotr Luszczek** (luszczek@icl.utk.edu).